Student Grade Book

CSN IS101

100 pts

Unit 3 — Excel Spreadsheets · Variation Project Option A

/ 100 points · 40 tasks

How to use this instruction sheet: Download the starter file from the Project Grader, open it in Microsoft Excel 365, and keep this sheet beside you. Work through each task in order. Each task tells you what to do, why it matters, where to find it, and the correct formula syntax — but you must enter the formula yourself. The starter file contains clean data only — no instructions are embedded in the file itself.

E1 — Manage Worksheets & Workbooks

18 points (8 tasks)

2

Task 1 — Add a new worksheet to the workbook and rename it "Grade Summary"

WHY THIS MATTERS

Organizing a workbook with multiple named sheets is a professional standard. Each sheet serves a distinct purpose and is clearly labeled.

WHERE TO FIND IT

Right-click any sheet tab at the bottom of Excel → Insert → Worksheet. Then double-click the new tab and type "Grade Summary". Press Enter.

HOW YOU'LL KNOW IT WORKED

A new sheet tab labeled "Grade Summary" appears at the bottom of the workbook.

2

Task 2 — Freeze row 2 and column A so student names stay visible while scrolling right

WHY THIS MATTERS

Grade books have many assignment columns. Freezing keeps student names visible as you scroll.

WHERE TO FIND IT

Click cell D3. View → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

Freeze lines appear. Scroll right — First Name, Last Name, Section stay visible.

2

Task 3 — Navigate to the named range "FinalScores" using the Name Box

WHY THIS MATTERS

Named ranges let you jump to important data instantly and confirm the range is defined correctly.

WHERE TO FIND IT

Click the Name Box → type FinalScores → press Enter.

HOW YOU'LL KNOW IT WORKED

The average score cells for all students are selected.

2

Task 4 — Set print area to the grade book; configure to print gridlines and row/column headers

WHY THIS MATTERS

Printing gridlines and headers helps graders track which cell they're reading.

WHERE TO FIND IT

Select the grade book range. Page Layout → Print Area → Set Print Area. Then Page Layout → Page Setup → Sheet tab → check Gridlines and Row and column headings.

HOW YOU'LL KNOW IT WORKED

Print Preview shows cell borders and column letters along the top.

2

Task 5 — Hide the NSHE ID column; adjust column widths for all assignment columns

WHY THIS MATTERS

NSHE IDs are sensitive data that should not appear in printed grade reports.

WHERE TO FIND IT

Right-click the NSHE ID column → Hide. Select assignment columns → right-click → Column Width → type a narrower value.

HOW YOU'LL KNOW IT WORKED

The NSHE ID column is not visible. Assignment columns are narrow and uniform.

2

Task 6 — Apply the "Wisp" workbook theme

WHY THIS MATTERS

The Wisp theme uses soft academic-appropriate colors suitable for educational documents.

WHERE TO FIND IT

Page Layout tab → Themes → find and click Wisp.

HOW YOU'LL KNOW IT WORKED

The workbook fonts and colors update to the Wisp theme.

2

Task 7 — Set workbook properties: Title = "IS101 Grade Book Spring 2026", Author = your name

WHY THIS MATTERS

Grade books must be clearly identified with course, term, and instructor for records retention.

WHERE TO FIND IT

File → Info → Properties panel → fill in Title and Author.

HOW YOU'LL KNOW IT WORKED

File → Info shows the correct Title and Author.

4

Task 8 — Save the completed grade book as PDF for records

WHY THIS MATTERS

PDF grade records are required for academic records retention and cannot be accidentally edited.

WHERE TO FIND IT

File → Save As → change type to PDF → Save.

HOW YOU'LL KNOW IT WORKED

A PDF version of the grade book exists in your folder.

E2 — Manage Data Cells & Ranges

18 points (8 tasks)

2

Task 9 — Use AutoFill to extend "Assignment 1" and "Assignment 2" through "Assignment 8"

WHY THIS MATTERS

AutoFill recognizes the Assignment 1, Assignment 2 pattern and continues it automatically.

WHERE TO FIND IT

The starter has "Assignment 1" in D2 and "Assignment 2" in E2. Select both. Drag fill handle right through column K.

HOW YOU'LL KNOW IT WORKED

Cells D2 through K2 show Assignment 1 through Assignment 8 in sequence.

2

Task 10 — Insert a new "Extra Credit" column between Midterm and Final

WHY THIS MATTERS

Extra credit is a separate grade component that needs its own column.

WHERE TO FIND IT

Right-click the Final column header → Insert. Label the new column "Extra Credit".

HOW YOU'LL KNOW IT WORKED

An "Extra Credit" column appears between Midterm and Final.

2

Task 11 — Format the Average % column as Percentage with 1 decimal place

WHY THIS MATTERS

Percentage formatting converts 0.923 to 92.3% — the format students and instructors expect.

WHERE TO FIND IT

Select the Average % column. Home → % button. Click increase decimals once for 1 decimal.

HOW YOU'LL KNOW IT WORKED

Average values show as 92.3% instead of 0.923.

2

Task 12 — Merge and center the title "CSN IS101 — Spring 2026 Grade Book" across the top row

WHY THIS MATTERS

A merged title makes the grade book immediately identifiable when printed or shared.

WHERE TO FIND IT

Select row 1 across all columns. Home → Merge & Center. Type the title.

HOW YOU'LL KNOW IT WORKED

Row 1 is one wide cell with the grade book title centered.

2

Task 13 — Right-align all score columns; center the Letter Grade column

WHY THIS MATTERS

Numbers right-aligned so decimal points line up. Letter grades centered in their column looks professional.

WHERE TO FIND IT

Select score columns → Home → Right Align. Select Letter Grade column → Home → Center.

HOW YOU'LL KNOW IT WORKED

Score columns are right-aligned. Letter grades are centered.

3

Task 14 — Apply conditional formatting: A grades green, B grades blue, C grades yellow, D/F grades red

WHY THIS MATTERS

Color-coded grades let instructors instantly spot failing students without reading every cell.

WHERE TO FIND IT

Select the Letter Grade column. Home → Conditional Formatting → Highlight Cells Rules → Equal To. Create 5 rules: A=green, B=blue, C=yellow, D=orange, F=red.

HOW YOU'LL KNOW IT WORKED

Each letter grade cell shows the correct background color.

2

Task 15 — Use Format Painter to copy the header row formatting to the summary row

WHY THIS MATTERS

Consistent formatting creates a visually balanced grade book.

WHERE TO FIND IT

Click a header row cell. Home → Format Painter (double-click to lock). Click across the summary row. Press Escape.

HOW YOU'LL KNOW IT WORKED

The summary row has the same bold, background, and borders as the header row.

3

Task 16 — Name the final score range "FinalScores" using Formulas → Define Name

WHY THIS MATTERS

Naming the range enables readable formulas like =MAX(FinalScores) throughout the grade book.

WHERE TO FIND IT

Select the Average % cells for all students. Formulas tab → Defined Names → Define Name → type FinalScores → OK.

HOW YOU'LL KNOW IT WORKED

Formulas tab → Name Manager shows FinalScores pointing to the Average % column.

E3 — Manage Tables & Table Data

14 points (6 tasks)

2

Task 17 — Convert the grade scale reference data on the Grade Scale sheet to an Excel Table

WHY THIS MATTERS

Converting to a Table makes VLOOKUP references more reliable and the scale easier to update.

WHERE TO FIND IT

Click inside the grade scale data on the Grade Scale sheet. Insert → Table → OK.

HOW YOU'LL KNOW IT WORKED

The grade scale has alternating colors and filter arrows in the header row.

2

Task 18 — Apply a gold Medium table style to the grade scale table

WHY THIS MATTERS

Gold distinguishes the grade scale reference table from the main grade book data.

WHERE TO FIND IT

Click inside the grade scale table. Table Design → Table Styles gallery → find a gold Medium style → click it.

HOW YOU'LL KNOW IT WORKED

The grade scale table uses a gold color scheme.

2

Task 19 — Add a Total Row to the assignment scores table showing class averages

WHY THIS MATTERS

Class averages tell the instructor whether each assignment was appropriately difficult.

WHERE TO FIND IT

Click inside the grade book table. Table Design → Total Row checkbox. Set each assignment column to Average.

HOW YOU'LL KNOW IT WORKED

A Total row at the bottom shows the class average for each assignment.

2

Task 20 — Filter the grade book to show only students who scored below 70% (failing)

WHY THIS MATTERS

Filtering for failing students lets an instructor quickly identify who needs intervention.

WHERE TO FIND IT

Click the Average % column dropdown → Number Filters → Less Than → type 0.7 → OK.

HOW YOU'LL KNOW IT WORKED

Only students with average below 70% are visible.

2

Task 21 — Sort the full grade book by Final Grade from highest to lowest

WHY THIS MATTERS

Sorting by grade shows class rank — useful for identifying top performers.

WHERE TO FIND IT

Click the Average % column dropdown → Sort Largest to Smallest.

HOW YOU'LL KNOW IT WORKED

The grade book rows are reordered with the highest-scoring student at the top.

4

Task 22 — Remove any duplicate student rows from the roster

WHY THIS MATTERS

Duplicate student records would double-count that student in all class averages.

WHERE TO FIND IT

Click inside the table. Table Design → Remove Duplicates → select all columns → OK.

HOW YOU'LL KNOW IT WORKED

Excel confirms 1 duplicate removed. The red duplicate row is gone.

E4 — Formulas & Functions

32 points (12 tasks)

2

Task 23 — Write =SUM(D3:K3) to total each student's assignment scores

WHY THIS MATTERS

SUM totals all assignment points for each student — the raw total before averaging.

WHERE TO FIND IT

Click the first Total cell (row 3). Copy down for all 30 students.

FORMULA SYNTAX

```
=SUM(D3:K3)
```

HOW YOU'LL KNOW IT WORKED

Each student's total shows the sum of their 8 assignment scores.

2

Task 24 — Write =AVERAGE(D3:K3) to calculate each student's average score

WHY THIS MATTERS

AVERAGE calculates the mean across all assignments giving an equal-weighted average.

WHERE TO FIND IT

Click the first Average cell. Copy down for all 30 students. Format as % with 1 decimal.

FORMULA SYNTAX

```
=AVERAGE(D3:K3)/100
```

HOW YOU'LL KNOW IT WORKED

Each student shows an average formatted as a percentage (e.g., 84.7%).

2

Task 25 — Write =MAX(FinalScores) for highest class grade and =MIN(FinalScores) for lowest

WHY THIS MATTERS

MAX and MIN show the class grade range between top and bottom performers.

WHERE TO FIND IT

Click Highest Grade cell → MAX formula. Click Lowest Grade cell → MIN formula.

FORMULA SYNTAX

```
=MAX(FinalScores) and =MIN(FinalScores)
```

HOW YOU'LL KNOW IT WORKED

Highest and lowest student averages appear in the summary.

2

Task 26 — Write =COUNT(FinalScores) to count students with recorded grades

WHY THIS MATTERS

COUNT verifies all students have a grade entered. If less than 30, some grades are missing.

WHERE TO FIND IT

Click the Students Recorded cell.

FORMULA SYNTAX

```
=COUNT(FinalScores)
```

HOW YOU'LL KNOW IT WORKED

The cell shows 30.

3

Task 27 — Write IF: if Final Average >= 70 show "Pass", else "Fail"

WHY THIS MATTERS

IF automates the pass/fail determination for all 30 students instantly.

WHERE TO FIND IT

Click the first Pass/Fail cell. Copy down for all students.

FORMULA SYNTAX

```
=IF(AVERAGE(D3:K3)>=70,"Pass","Fail")
```

HOW YOU'LL KNOW IT WORKED

Students with 70%+ show Pass. Students below 70% show Fail.

4

Task 28 — Write nested IF to assign letter grades: A(90+), B(80+), C(70+), D(60+), F

WHY THIS MATTERS

Nested IFs evaluate multiple conditions in sequence — the most common use in educational grading.

WHERE TO FIND IT

Click the first Letter Grade cell. Copy down for all 30 students.

FORMULA SYNTAX

```
=IF(AVERAGE(D3:K3)>=90,"A",IF(AVERAGE(D3:K3)>=80,"B",IF(AVERAGE(D3:K3)>=70,"C","D","F"))))
```

HOW YOU'LL KNOW IT WORKED

Each student shows the correct letter grade matching their average.

3

Task 29 — Write COUNTIF to count how many students earned each letter grade

WHY THIS MATTERS

Grade distribution tells the instructor whether the assessment was appropriately difficult.

WHERE TO FIND IT

Click each grade count cell in the summary. Repeat for A, B, C, D, F.

FORMULA SYNTAX

```
=COUNTIF(Q3:Q32,"A")
```

HOW YOU'LL KNOW IT WORKED

The five count cells add up to 30.

3

Task 30 — Write SUMIF to total points earned by Section A and Section B students separately

WHY THIS MATTERS

SUMIF totals values meeting a condition. Section comparison helps identify performance differences.

WHERE TO FIND IT

Click the Section A total cell and Section B total cell.

FORMULA SYNTAX

```
=SUMIF(C3:C32,"A",O3:O32)
```

HOW YOU'LL KNOW IT WORKED

Section A and B totals together equal the grand total of all points.

2

Task 31 — Use TEXTJOIN to combine First Name and Last Name into a Full Name column

WHY THIS MATTERS

TEXTJOIN joins multiple text values with a delimiter — combining split name columns is a common data prep task.

WHERE TO FIND IT

Click the first Full Name cell (column S). Copy down for all students.

FORMULA SYNTAX

```
=TEXTJOIN(" ",TRUE,A3,B3)
```

HOW YOU'LL KNOW IT WORKED

Each Full Name cell shows "FirstName LastName" as one combined string.

2

Task 32 — Use TODAY() to display the date the grade book was last updated

WHY THIS MATTERS

A timestamp shows when the grade book was last modified — important for version control.

WHERE TO FIND IT

Click the Last Updated cell in the summary area.

FORMULA SYNTAX

```
=TODAY()
```

HOW YOU'LL KNOW IT WORKED

The cell shows today's date and updates automatically each day.

4

Task 33 — Write VLOOKUP to pull the grade scale description from the Grade Scale sheet

WHY THIS MATTERS

VLOOKUP looks up each student's score in the grade scale table and returns the description automatically.

WHERE TO FIND IT

Click the first Grade Description cell.

FORMULA SYNTAX

```
=IFERROR(VLOOKUP(ROUND(AVERAGE(D3:K3),0),'Grade Scale'!$A$3:$C$7,3,TRUE),"Check Score")
```

HOW YOU'LL KNOW IT WORKED

Each student shows the correct grade description matching their average.

3

Task 34 — Wrap the VLOOKUP in IFERROR to show "Check Score" if grade is missing

WHY THIS MATTERS

IFERROR handles blank scores or out-of-range values, displaying a helpful message instead of an error.

WHERE TO FIND IT

The IFERROR is already included in the formula above. Verify it works for any blank score cells.

FORMULA SYNTAX

```
=IFERROR(VLOOKUP(...),"Check Score")
```

HOW YOU'LL KNOW IT WORKED

Students with missing scores show "Check Score" instead of #N/A.

E5 — Manage Charts

18 points (6 tasks)

3

Task 35 — Create a Column chart showing grade distribution (count of A, B, C, D, F)

WHY THIS MATTERS

A grade distribution chart shows the class performance curve — essential for course assessment reporting.

WHERE TO FIND IT

Select grade letters (A-F) and their counts from the summary area. Insert → Column or Bar Chart → Clustered Column.

HOW YOU'LL KNOW IT WORKED

A column chart appears with 5 bars showing how many students earned each grade.

2

Task 36 — Move the grade distribution chart to its own sheet named "Grade Distribution"

WHY THIS MATTERS

A dedicated sheet displays the chart at full size for course assessment reports.

WHERE TO FIND IT

Click chart. Chart Design → Move Chart → New sheet → type "Grade Distribution" → OK.

HOW YOU'LL KNOW IT WORKED

A "Grade Distribution" sheet tab appears with only the chart.

3

Task 37 — Change the assignment average chart type from Column to Line

WHY THIS MATTERS

Line charts show trends across a series — making it easy to see if students improved over the semester.

WHERE TO FIND IT

Click the assignment averages chart. Chart Design → Change Chart Type → Line → OK.

HOW YOU'LL KNOW IT WORKED

The chart changes from bars to a connected line.

3

Task 38 — Add chart title "Class Grade Distribution" and data labels to the column chart

WHY THIS MATTERS

Data labels show exact counts on each bar so readers see numbers without estimating.

WHERE TO FIND IT

Click chart. Chart Design → Add Chart Element → Chart Title → type title. Then Add Chart Element → Data Labels → Outside End.

HOW YOU'LL KNOW IT WORKED

The chart has a title and each bar shows its count above it.

3

Task 39 — Apply Quick Layout 3 and a color chart style to the distribution chart

WHY THIS MATTERS

Quick Layout 3 adds a legend. A color style makes grade categories visually distinct.

WHERE TO FIND IT

Click chart. Chart Design → Quick Layout → Layout 3. Then Chart Design → Chart Styles → choose a colorful style.

HOW YOU'LL KNOW IT WORKED

The chart uses Quick Layout 3 with a colorful style applied.

4

Task 40 — Add alt text to all charts in the workbook

WHY THIS MATTERS

Accessible grade books are required under Section 508 and ADA guidelines in educational settings.

WHERE TO FIND IT

Right-click each chart → Edit Alt Text → describe what the chart shows.

HOW YOU'LL KNOW IT WORKED

Every chart has a descriptive alt text entry.